

# ATRI SUKUL

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## SUMMARY

Machine Learning Practitioner working on healthcare AI at IIT Madras, where I am building clinical NLP systems to make medical information more accessible and useful. I enjoy understanding ideas from research papers and putting them into practice, and have co-authored research published in IEEE TGRS and the Proceedings of the CVPR Conference.

## EDUCATION

**Indian Institute of Technology, Tirupati**

M.Tech in Data Science and Artificial Intelligence | CGPA: 8.69

07/2025 – Present

Tirupati, India

**Jalpaiguri Government Engineering College**

B.Tech in Computer Science and Engineering | CGPA: 9.33

08/2021 – 06/2025

Jalpaiguri, India

## EXPERIENCE

**Project Associate | IC&SR, IIT Madras (with UHG Optum)**

08/2025 – Present

Healthcare AI / NLP

Chennai, India (Remote)

- Building an NLP pipeline that transforms unstructured EHR discharge reports into knowledge graphs, making relevant clinical information easier to identify and use.
- Designing a context-grounded GraphRAG question-answering system for clinical records to support faster and more reliable information retrieval.

**Student Fellow | Indian National Academy of Engineering (INAE)**

02/2024 – 03/2024

Mentoring of Engineering Students by INAE Fellow

India

- Selected under the “Mentoring of Engineering Students by INAE Fellow-2021” scheme, one of only 60 students chosen across India that year for the fellowship.
- Mentored by Prof. Umapada Pal, ISI Kolkata, under the scheme (2023–24); built a frequency-domain generative framework for hyperspectral image classification, reaching 95.52% (Pavia) and 99.05% (Salinas) and surpassing models such as SpectralDiff.

## SKILLS

**Technologies:** PyTorch, LangChain, FastAPI, OpenCV, Pandas, NumPy, Scikit-Learn, ReactJS, NodeJS

**Languages:** Python, SQL, C, C++, JavaScript

**Databases:** MySQL, MongoDB

**Specializations:** LLM, NLP, Computer Vision, Deep Learning, Remote Sensing

## PROJECTS

**clawss package | FastAPI, React, PyTorch, Ollama/OpenRouter | GitHub**

05/2026

- A browser-based modular notebook where cells behave like Python files with isolated namespaces and cross-cell imports; adds a cell-dependency graph, PyTorch architecture visualization, and notebook-aware AI generation via local or API-based models.

**Designed a fingerprint recognition system using few-shot learning | PyTorch**

01/2025

- Developed a few-shot fingerprint recognition system using Siamese Networks on the SOCOFing dataset.

**Cross Hyperspectral and LiDAR Attention Transformer | PyTorch | GitHub**

08/2023

- Implemented the full codebase for the IEEE TGRS paper, realizing the proposed extended self-attention method for land-use and land-cover classification.

## PUBLICATIONS

- S. K. Roy, **Atri Sukul**, A. Jamali, J. Haut, P. Ghamisi. “Cross Hyperspectral and LiDAR Attention Transformer: An Extended Self-Attention for Land Use and Land Cover Classification.” *IEEE Transactions on Geoscience & Remote Sensing*, 2024. [IEEE Xplore]
- S. Mitra, **Atri Sukul**, S. K. Roy, P. Singh, V. K. Verma. “ScoreCAM++: Gated Score-Weighted Visual Explanations for CNNs.” *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2700-2709. [CVPR Open Access]

## ACHIEVEMENTS

- Smart India Hackathon (SIH) 2022 Winner** | Team Odyssey (PS - Virtual Excavation Game). [Link]
- Hack4Bengal 2.0** | Secured the 1st Runner-Up spot for the Flow Track in Hack4Bengal 2.0 [Link]

## LEADERSHIP AND VOLUNTEERING

- AI/ML Lead**, Google Developer Student Club, JGEC (2023–2024).
- Deputy Secretary**, National Service Scheme (NSS) Unit, JGEC.
- Lead Organizer**, JWOC 2024 | a nationwide open-source program with 1900+ participants.